



The OverSlapper Documentation

A tool by Philippe Ratté

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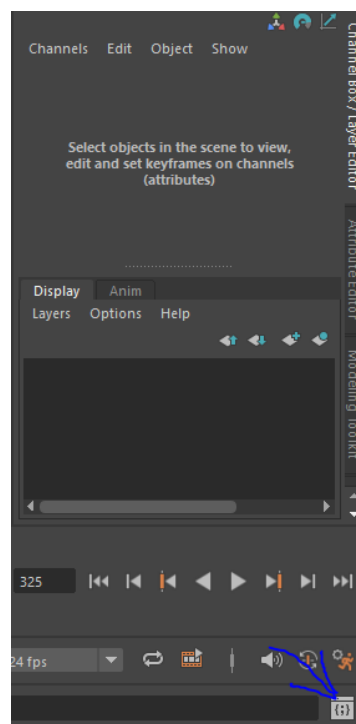
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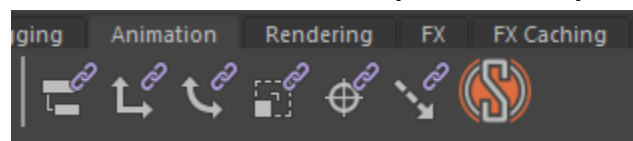
Installation

Installing the tool

1. Have the the overslapper zip downloaded
2. Copy the overslapper folder and the overslapper_installer.py to this location. If any one of these paths does not correspond to your maya path please run the overslapper_installer.py in your script editor and it will print the path where you need to put the files.
 - For windows (replace <username> by your windows username)
 - \Users\<username>\Documents\Maya\scripts
 - For macOS
 - \$HOME/Library/Preferences/Autodesk/Maya/scripts
 - For linux machine
 - \$HOME/Maya/scripts
3. Then open and run the overslapper_installer.py in the script editor the button for the tool will be installed on the currently active shelf



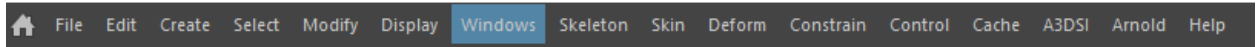
4. Then the tool is now installed and you are ready to start overlapping



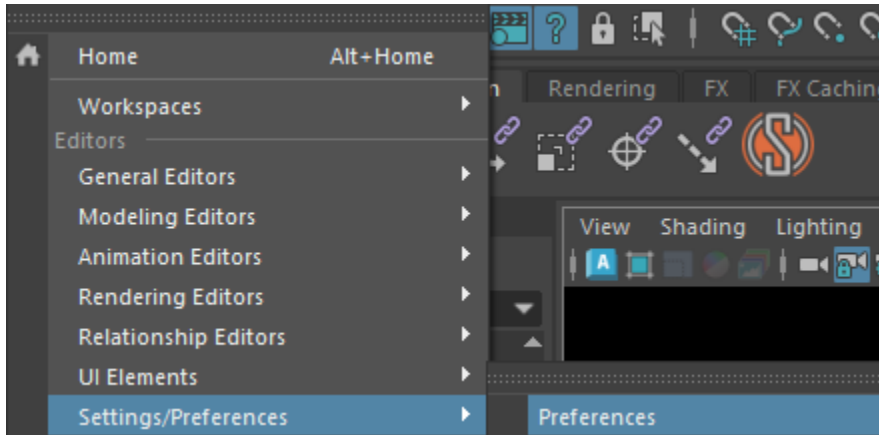
Activating tool tips

If you can't see the tool tips of the tool by default you will need to activate it in Maya

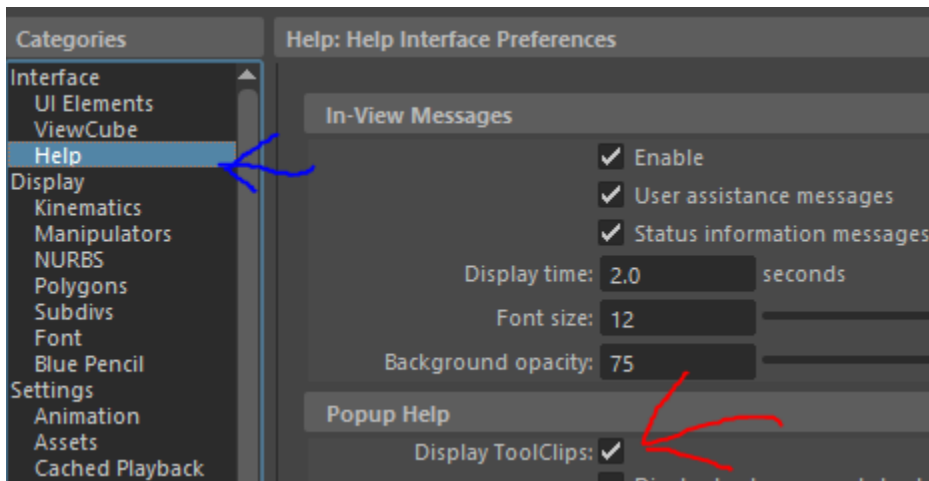
1. First go to the window tab



2. Go into window>Settings/Preferences>Preferences



3. Go into the help section and turn on the Display ToolClips: option

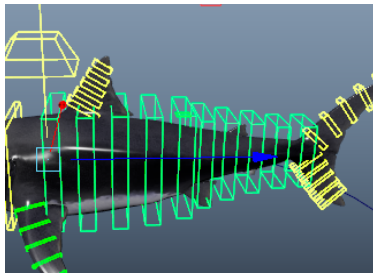


Rotational overlap:

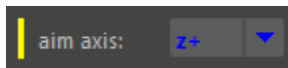


Aim axis:

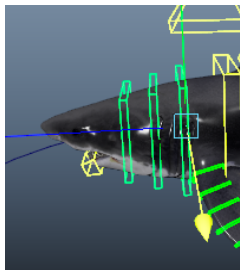
This setting needs to be set to the axis that is pointing to the next control



For example on this rig we can see that the shark spine control are point to each other in the z positive axis so we need to set the aim axis to it



Pro tip: let say that you are moving the shark forward in space or along a path and set the aim axis to z positive the head would flip so the solution to avoid it breaking would be to set the aim axis to z negative to avoid the flip.



Up axis:

This setting need to be set to the up axis of your current selection



For the same example the up axis should be y positive

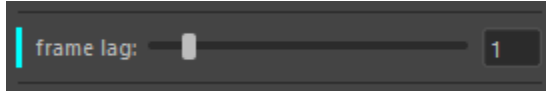


Translation overlap:



Frame lag:

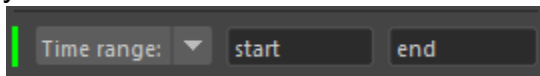
The frame lag attribute determine how many frame the translation will lag behind



Basic settings of the tool

Time range:

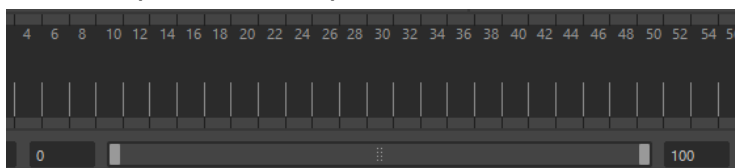
- There is three options for the time range
- The first one will let you input manually the start and the end of the overlap that you want to create



- The second one will query your current timeline and create the overlap from the star to the end of that time range



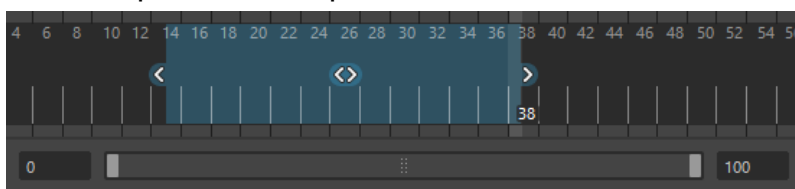
For example the overlap will be created from frame 0 to 100



- The third one will query the current selection in the timeline and create the overlap from it



For example the overlap will be created from frame 14 to 38



Strength:

- The strength attribute will multiply the values of the overlap at the end of the calculation



- You may input a negative multiplier and it will create an inverse version of your overlap

Stiffness:

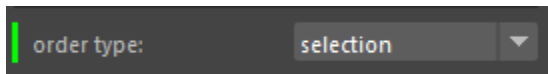
- The stiffness attribute will determine how stiff or how bendy your overlap will be



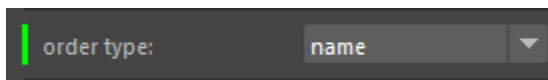
- A stiffness of 1 will be absolutely stiff and will not move
- A stiffness of 0 will be extremely bendy

Order:

- There are three options for the order attribute
- The first one is selection and the overlap will be created in the order that you have selected your controllers



- The second one is name and will order your selection you give it by name then create the overlap



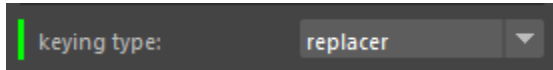
- The third one is split by name which will divide your selection in sections based on the name of your selection.



It is especially a powerful option if you are using the complex stiffness option because the complex stiffness will be applied on each of these sections instead of being created from the start of the selection to the end of it

Keying type:

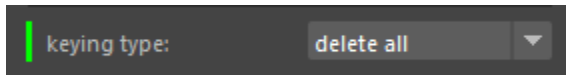
- There is three option for the keying type
- The first one is replacer which will replace all the keys for the transform you have selected and its selected axis in the time range you have imputed



- The second one is add and it will keep the keys you have on the transform and their axis for the time range and add them back into the overlap after calculating

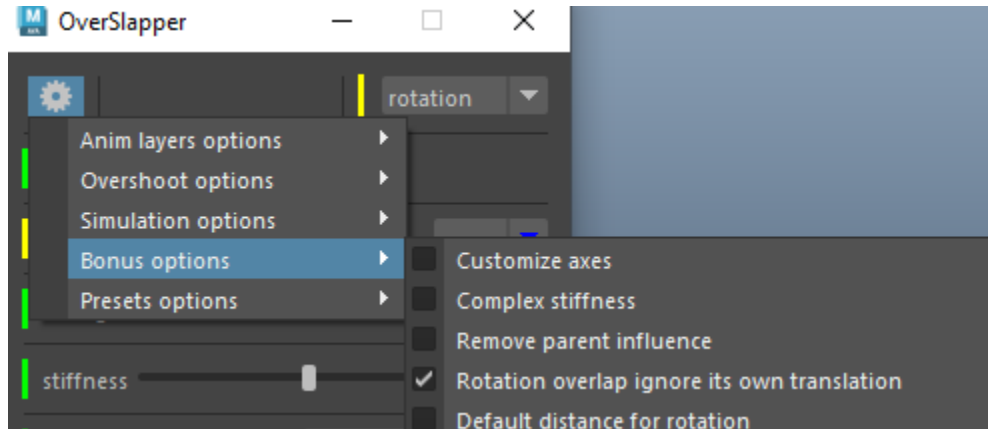


- The third one is delete all and it will delete every key you had on your selection for the transform you have selected and its selected axis across all of the animation graph. If you are in an anim layer it will only delete the keys on that anim layer.



Bonus options of the tool:

The bonus options can be found in the option menu that can be found by pressing the gear icon on the top left of the tool



There is five checkbox that will change the behavior of the tool

Customize axes:

When activating this option you will now be able to select on which axis the overlap and other options will be animated

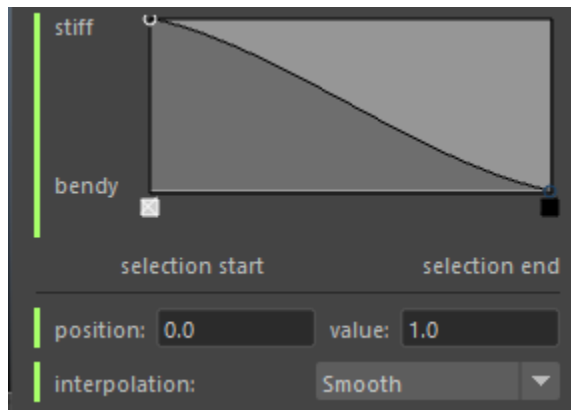


Complex stiffness:

The complex stiffness setting will replace the default stiffness attribute by a graph that you will determine the stiffness from the start to the end of the selection or from the start to the end of each splits if you are using split by name

To create new points on the graph just click on the graph at the position and value you want to set and then drag to modify it. You can manually change its position or value through the option below the graph. You can also change the interpolation of the point you have selected through the combo box at the end.

If the value dont seems to match the point you have selected just move it slightly and the tool will update those values



Remove parent influence:

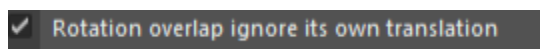
Checking this option will let you remove the influence of a parent on your overlap. You can either write manually the name of the parent you want to remove or if you select the parent then press the get button it will input it into the tool



Example of a use case for this option. You have a cycle that you move through the world with the SRT control but you only want to get the motion of your COG control for your overlap. Then what you would do is turn on the remove parent influence option, select the SRT control, press get, then run the overlapper and you will obtain an overlap as if the SRT control had never been transformed.

Rotation overlap ignore its own translation:

This option is ON by default and will make it so that rotational overlap will not calculate their own translation to create the overlap.

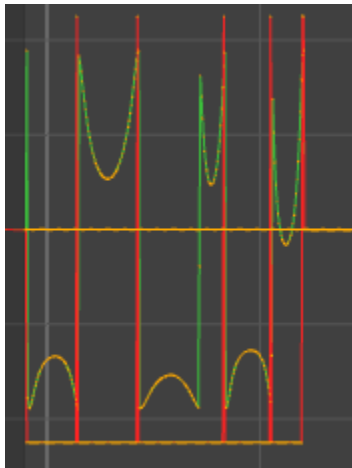


Default distance for rotation:

This option is there for in cases where you are selecting only one control for an overlap and the tool cannot find a child to determine the distance between controls. It is also good practice to use this option when using the split by name selection option because you can end up with only one object in the created splits.

In general the tool will calculate the distance between objects in the selection and use that distance to calculate the rotations.

If you are getting weird looking rotational curves like one of these its because the tool could not find a proper distance and set the default distance to 1



To set a default distance just select 2 controllers and press the get button

Default control distance:

then the default distance will be set to the distance between those 2 controllers.

Example

Default control distance:

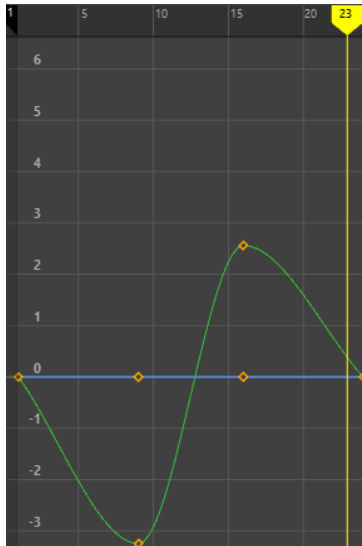
You can also turn on the option to only use the default distance and the tool will only use the value that you have set for all the controllers in your selection

Cycle animation:

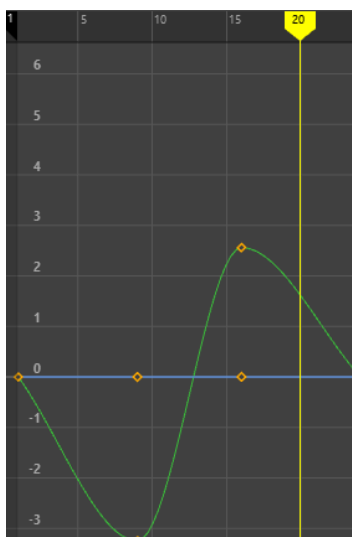
This option is off by default. Turning it on will make it so that the tool will interpret the driving animation as a cycle.



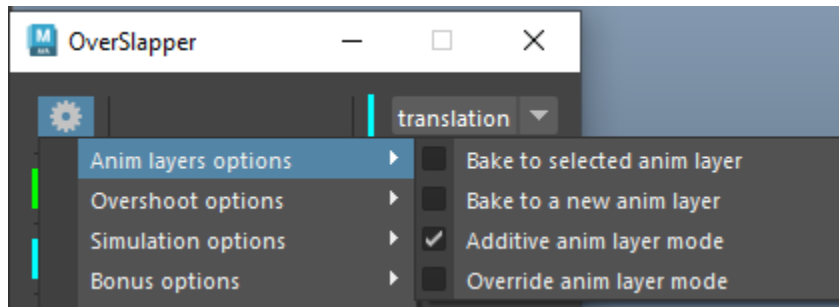
If its a rotation overlap the tool will look one frame before the last frame for the first frame



If its a translation overlap the tool will look at $x(\text{frame lag})$ frame before in the cycle.
Example of where the tool would look at for the first frame:



Anim layers option:



Bake to selected anim layer:

Will add the selected controllers to the topmost selected anim layer and will bake the overlap in that layer.

Bake to a new anim layer:

Will create an anim layer and will bake the overlap in that new anim layer.

Additive anim layer mode:

The anim layer that you will create with bake to a new anim layer will be set to the additive mode

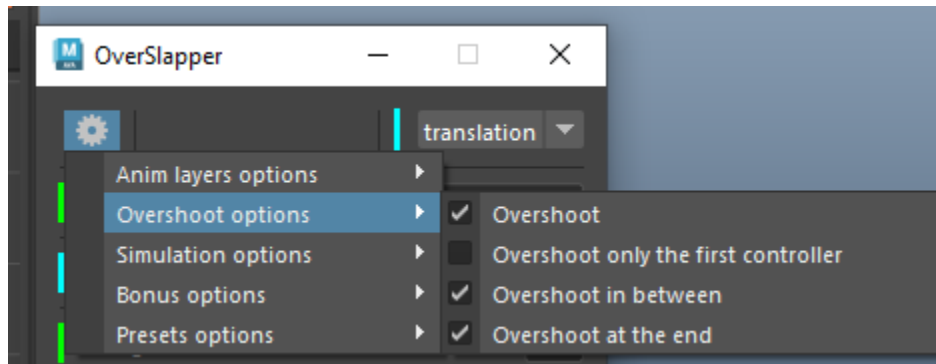
Override anim layer mode:

The anim layer that you will create with bake to a new anim layer will be set to the override mode

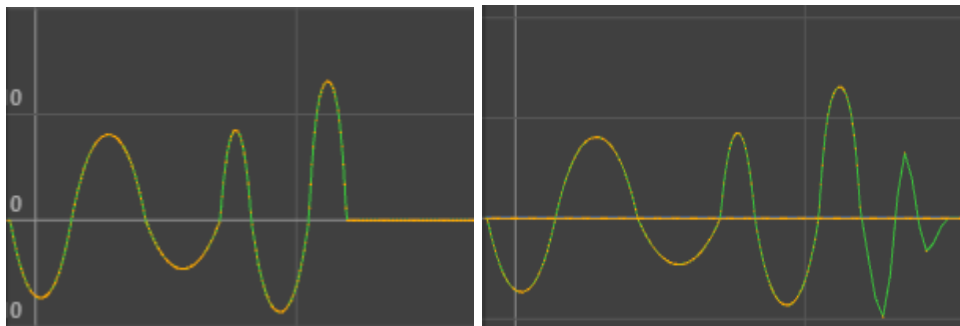
Warning for using The overslapper with anim layers:

With the way that anim layers work in maya, for each layer that you have the controllers that you are overlapping, you will reduce the speed at which the tool to create the keyframes for the overlap something that could take 17 seconds to overlap can take over a minute if the selected controllers are in 5 anim layers at the same time.

Overshoot option:



Example of an overlap curve without and with the overshoot option turned on



The overshoot will only create the keys at the peak and valleys to make it simpler to modify if the user wants to

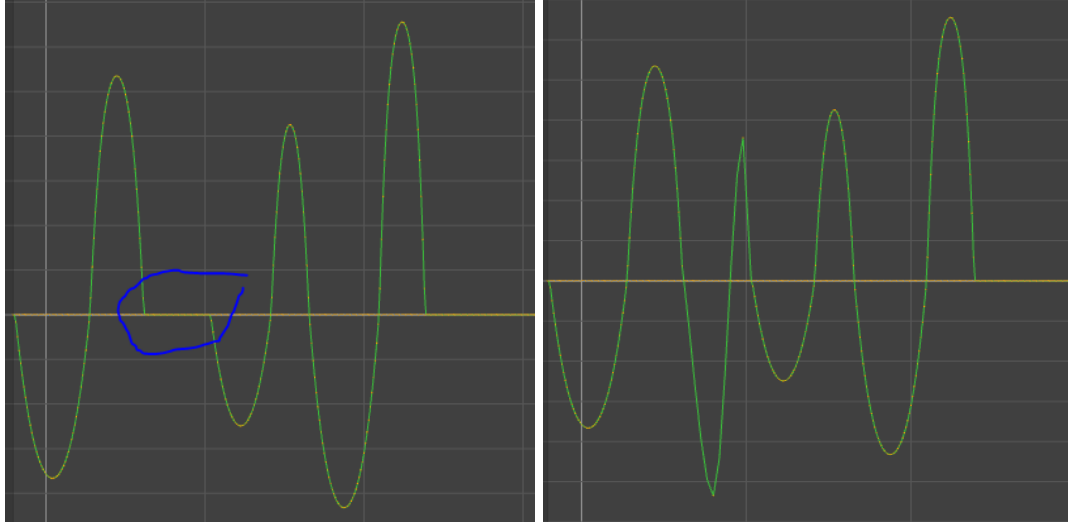
Overshoot only the first controller:

Will create the overshoot only on the first selected controller, the first controller in the name order or only on the first controller of each splits in split by name

Pro tip: This option is best to use for rotation overlap

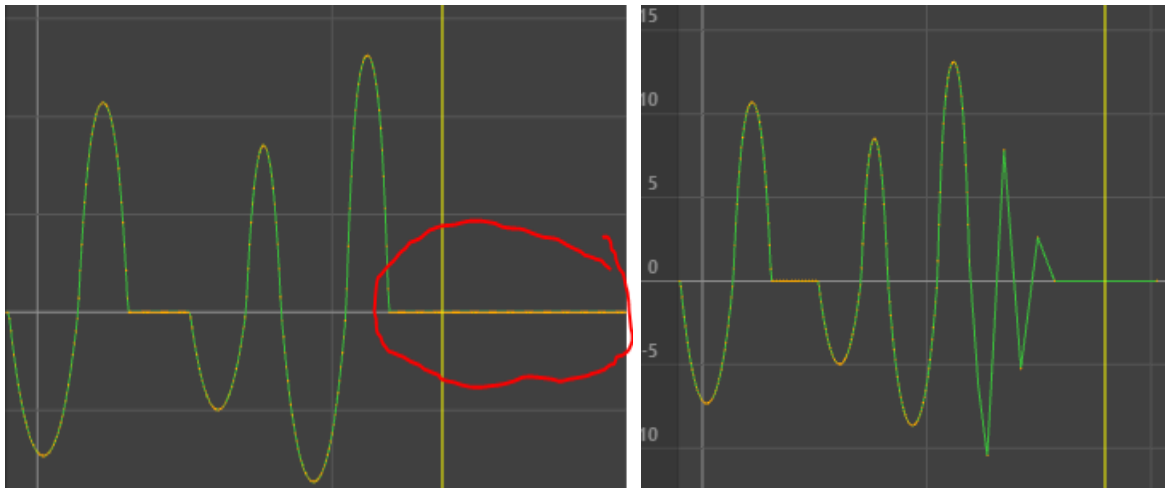
Overshoot in between:

This option will make it so that the tool will automatically create and overshoot in all stable zones before the last stable zone that is stable in your overlap. The overlap attributes will not affect these sections



Overshoot at the end:

This option will make it so that the tool will create overshoot values at the last stable zone in the animation. This section of the overshoot is affected by the overshoot attributes.



Overshoot attributes:

Overshoot strength:

This attribute will multiply the values for the keys of the overshoot at the last zone of stability

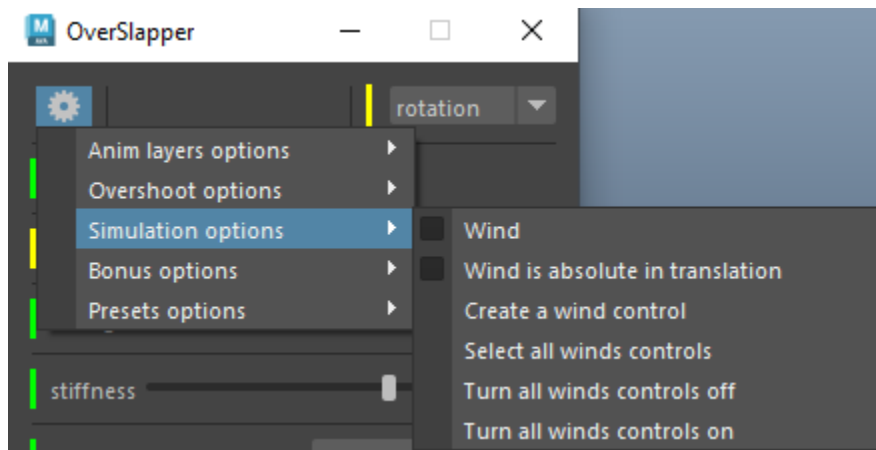


Oscillation number:

This attribute will set the number of time the overshoot will oscillate at the last zone of stability



Simulation option:



Wind option:

Turning on this option will evaluate all the currently active wind controls in the scene for the overlap

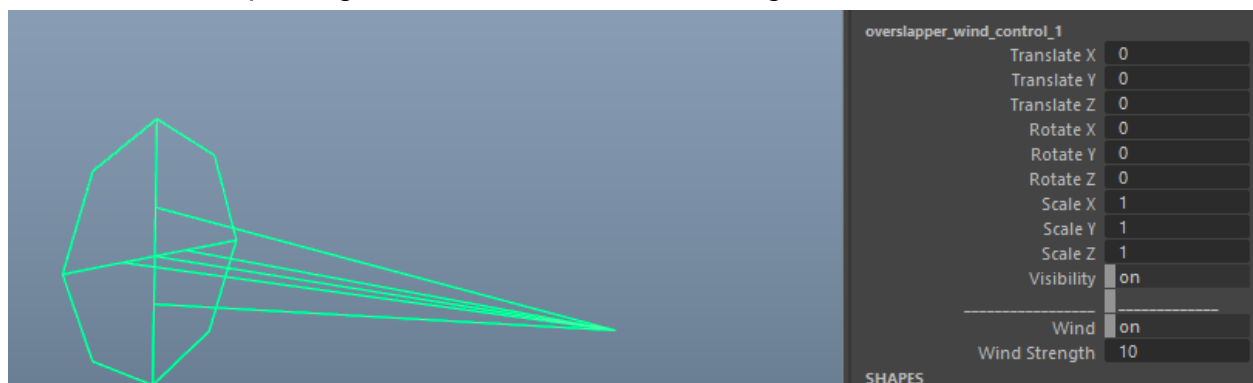
Wind is absolut in translation option:

Turning on this option will make it so that when you simulate wind for a translation overlap, the winds will push the controller to a new position in the world.

Create a wind control:

Will create a control in your maya scene that will drive the wind simulation.

- You can have as many wind controls as you want in your scene and they will all be evaluated if they are turned on.
- You can rename them without issue and the tool will be able to find them with no issue.
- I do not recommend duplicating them since it will not duplicate the connection from the wind strength attribute with the controller shapes.
- The wind direction is controlled with the rotation and is absolute in the world. The controller translation and scales do not affect the final wind result.
- You can turn a wind controller on and off through the wind attribute of the control. And it will change the color of the control from green to blue
- If a control is not visible its wind attribute is still on, it will still be evaluated for the wind.
- You can input negative values in the wind strength of the control



Select all winds controls:

Will select all winds controls that currently are in the scene

Turn all winds controls off:

Will turn all winds controls off in the scene

Turn all winds controls on:

Will turn all winds controls on in the scene

Wind attribute

Wind strength:

This attribute will multiply the values of all the winds control wind strength attribute by its own value



Preset options

Export a preset:

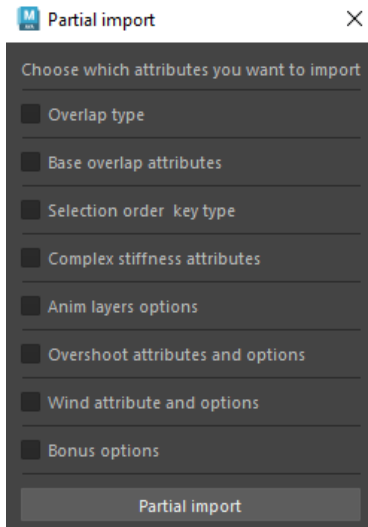
Will create a custom .json file that will save all the values and setting in the ui

Import a preset:

Will change the ui based on the values and setting for the ui in the .json that you chose

Partial import a preset:

Will let you pick and choose which settings and values you will import by categories



Default.json

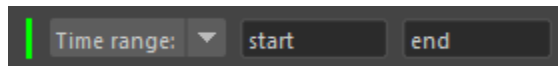
In the overlapper folder you will find a default.json file if you want you can overwrite it with your favorite options and values of the tool and each time you will open the tool that default.json will be loaded in as the new default form of the tool. If you ever get an error telling you that the default file is corrupted or is missing no issue the tool will launch in the standard mode and you can re export a default.json and the next time you launch the tool it will use that new default.

icons	2024-09-28 7:35 PM	File folder	
packages	2024-10-02 12:36 PM	File folder	
default	2024-10-13 11:56 AM	JSON File	2 KB
overlapper_tool	2024-10-13 11:54 AM	Python File	104 KB

Errors

Sometimes when using the tool it will tell you when you make an error using don't be scared. It's not a bug, it's just that you need to fix the settings that you are using based on what the error message is telling you.

For example, if you are using the time range option but don't input values it will give you an error.



Please input a value in the start of the time range

Please input a value in the end of the time range

Or another example would be if all your settings are good but you don't have anything selected.

you need to select something